

Final Project

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Read in the cogimp0 dataset

```
# Access and read the cogimp0 file from the drive
cogimp0 = read_sav("/Users/wumanning/Downloads/COGIMP9220A_R.sav")

# generate codebook dataframe
cogimp0_codebook = labelled::generate_dictionary(cogimp0)
```

Read in the 2022 tracker file dataset

```
# read in the trk2022v2 file
trk2022v2 = read_sav("/Users/wumanning/Downloads/trk2022tr_r.sav")

# generate codebook
trk2022v2_codebook = labelled::generate_dictionary(trk2022v2)
```

Combine the data and variables we need

```
cogimp0 <- cogimp0 %>% mutate(hhid = as.character(HHID), pn = as.character(PN))
trk2022v2 <- trk2022v2 %>% mutate(hhid = as.character(hhid), pn = as.character(pn))

data_w13 <- cogimp0 %>%
  dplyr::select(hhid, pn, R13MO, R13DY, R13YR, R13DW, R13SCIS, R13CACT, R13PRES, R13VP) %>% rename(
    Month= R13MO,
    Day= R13DY,
    Year= R13YR,
    DOW= R13DW,
    Scissors= R13SCIS,
    Cactus= R13CACT,
    President= R13PRES,
    VP = R13VP
  ) %>%
  mutate(wave = "W13")
```

```
cov_w13 <- trk2022v2 %>%
  dplyr::select(hhid, pn, page, gender, degree, race, immgyear, usborn, hispanic) %>%
  rename(age = page) %>%
  mutate(wave = "W13")
```

```
lca_raw <- data_w13 %>%
  inner_join(cov_w13, by = c("hhid", "pn", "wave"))

indicator_names <- c("Month", "Day", "Year", "DOW", "Scissors", "Cactus", "President", "VP")
```

```
lca_ready <- lca_raw %>%
  dplyr::mutate(across(all_of(indicator_names),
    ~ ifelse(is.na(.), NA, as.integer(. + 1))))
```

```
summary(lca_raw[, indicator_names])
```

```
##      Month      Day      Year      DOW
##  Min.   :0.000   Min.   :0.000   Min.   :0.000   Min.   :0.000
##  1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000   1st Qu.:1.000
##  Median :1.000   Median :1.000   Median :1.000   Median :1.000
##  Mean   :0.959   Mean   :0.808   Mean   :0.958   Mean   :0.958
##  3rd Qu.:1.000   3rd Qu.:1.000   3rd Qu.:1.000   3rd Qu.:1.000
##  Max.   :1.000   Max.   :1.000   Max.   :1.000   Max.   :1.000
##  NA's   :27341   NA's   :27341   NA's   :27341   NA's   :27341
##  Scissors  Cactus  President  VP
##  Min.   :0.000   Min.   :0.00   Min.   :0.000   Min.   :0.00
##  1st Qu.:1.000   1st Qu.:1.00   1st Qu.:1.000   1st Qu.:0.00
##  Median :1.000   Median :1.00   Median :1.000   Median :1.00
##  Mean   :0.985   Mean   :0.92   Mean   :0.972   Mean   :0.63
##  3rd Qu.:1.000   3rd Qu.:1.00   3rd Qu.:1.000   3rd Qu.:1.00
##  Max.   :1.000   Max.   :1.00   Max.   :1.000   Max.   :1.00
##  NA's   :27341   NA's   :27341   NA's   :27341   NA's   :27341
```

```
summary(lca_raw$age)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      21.0   64.0   92.0   526.1   999.0   999.0
```

```
table(lca_raw$gender, useNA = "ifany")
```

```
##
##      1      2
## 17830 23211
```

```
table(lca_raw$degree, useNA = "ifany")
```

```
##
##      0      1      2      3      4      5      6      9
## 10117 2015 17600 2041 4771 2418 797 1282
```

```
table(lca_raw$race, useNA = "ifany")
```

```
##
##      0      1      2      7
## 111 29627 7871 3432
```

```
table(lca_raw$wave)
```

```
##
## W13
## 41041
```

Fit LCA model

Without covariates

```
set.seed(687)
f_ind <- cbind(Month, Day, Year, DOW, Scissors, Cactus, President, VP) ~ 1
sub_df <- lca_ready %>% dplyr::sample_frac(0.5)
fit_list <- lapply(2:5, function(k) {
  polCA(f_ind, data = sub_df, nclass = k, maxiter = 1500, nrep = 3, verbose = FALSE)
})

data.frame(
  k = 2:5,
  AIC = sapply(fit_list, \ (m) m$aic),
  BIC = sapply(fit_list, \ (m) m$bic),
  Gsq = sapply(fit_list, \ (m) m$Gsq)
)
```

```
## k      AIC      BIC      Gsq
## 1 2 27524.67 27640.81 693.3718
## 2 3 27199.26 27376.89 349.9661
## 3 4 27115.02 27354.14 247.7247
## 4 5 27081.23 27381.83 195.9287
```

After comparing AIC and BIC and considering simplicity and efficient, we chose 4 classes due to the lowest BIC and a relatively low AIC.

```
set.seed(687)
lca_k4 <- polCA(f_ind, data = lca_ready, nclass = 4, maxiter = 5000, nrep = 20, verbose = TRUE)
```

```
## Model 1: llik = -26936.26 ... best llik = -26936.26
## Model 2: llik = -26936.26 ... best llik = -26936.26
## Model 3: llik = -26936.26 ... best llik = -26936.26
## Model 4: llik = -26936.26 ... best llik = -26936.26
## Model 5: llik = -26936.26 ... best llik = -26936.26
## Model 6: llik = -26936.26 ... best llik = -26936.26
## Model 7: llik = -26936.26 ... best llik = -26936.26
```

```

## Model 8: llik = -26936.26 ... best llik = -26936.26
## Model 9: llik = -26936.26 ... best llik = -26936.26
## Model 10: llik = -26936.26 ... best llik = -26936.26
## Model 11: llik = -26936.26 ... best llik = -26936.26
## Model 12: llik = -26936.26 ... best llik = -26936.26
## Model 13: llik = -26936.26 ... best llik = -26936.26
## Model 14: llik = -26936.26 ... best llik = -26936.26
## Model 15: llik = -26936.26 ... best llik = -26936.26
## Model 16: llik = -26936.26 ... best llik = -26936.26
## Model 17: llik = -26936.26 ... best llik = -26936.26
## Model 18: llik = -26936.26 ... best llik = -26936.26
## Model 19: llik = -26936.26 ... best llik = -26936.26
## Model 20: llik = -26936.26 ... best llik = -26936.26
## Conditional item response (column) probabilities,
## by outcome variable, for each class (row)
##
## $Month
##           Pr(1) Pr(2)
## class 1: 0.0102 0.9898
## class 2: 0.7043 0.2957
## class 3: 0.3854 0.6146
## class 4: 0.0356 0.9644
##
## $Day
##           Pr(1) Pr(2)
## class 1: 0.1261 0.8739
## class 2: 0.9322 0.0678
## class 3: 0.7832 0.2168
## class 4: 0.2831 0.7169
##
## $Year
##           Pr(1) Pr(2)
## class 1: 0.0069 0.9931
## class 2: 0.8625 0.1375
## class 3: 0.3461 0.6539
## class 4: 0.0463 0.9537
##
## $DOW
##           Pr(1) Pr(2)
## class 1: 0.0113 0.9887
## class 2: 0.6371 0.3629
## class 3: 0.3049 0.6951
## class 4: 0.0624 0.9376
##
## $Scissors
##           Pr(1) Pr(2)
## class 1: 0.0058 0.9942
## class 2: 0.1112 0.8888
## class 3: 0.0132 0.9868
## class 4: 0.0476 0.9524
##
## $Cactus
##           Pr(1) Pr(2)
## class 1: 0.0253 0.9747

```

```

## class 2: 0.4581 0.5419
## class 3: 0.0613 0.9387
## class 4: 0.2969 0.7031
##
## $President
##           Pr(1) Pr(2)
## class 1: 0.0000 1.0000
## class 2: 0.6065 0.3935
## class 3: 0.0315 0.9685
## class 4: 0.0953 0.9047
##
## $VP
##           Pr(1) Pr(2)
## class 1: 0.2406 0.7594
## class 2: 1.0000 0.0000
## class 3: 0.5197 0.4803
## class 4: 0.8702 0.1298
##
## Estimated class population shares
## 0.7772 0.0183 0.0379 0.1665
##
## Predicted class memberships (by modal posterior prob.)
## 0.8693 0.0176 0.0315 0.0817
##
## =====
## Fit for 4 latent classes:
## =====
## number of observations: 13700
## number of estimated parameters: 35
## residual degrees of freedom: 220
## maximum log-likelihood: -26936.26
##
## AIC(4): 53942.53
## BIC(4): 54205.91
## G^2(4): 312.577 (Likelihood ratio/deviance statistic)
## X^2(4): 350.7069 (Chi-square goodness of fit)
##

```

```
lca_k4$probs
```

```

## $Month
##           Pr(1) Pr(2)
## class 1: 0.01021639 0.9897836
## class 2: 0.70425104 0.2957490
## class 3: 0.38540730 0.6145927
## class 4: 0.03557407 0.9644259
##
## $Day
##           Pr(1) Pr(2)
## class 1: 0.1261458 0.87385418
## class 2: 0.9322472 0.06775285
## class 3: 0.7831913 0.21680871
## class 4: 0.2830916 0.71690836
##

```

```

## $Year
##           Pr(1)      Pr(2)
## class 1: 0.00694969 0.9930503
## class 2: 0.86246217 0.1375378
## class 3: 0.34613287 0.6538671
## class 4: 0.04630827 0.9536917
##
## $DOW
##           Pr(1)      Pr(2)
## class 1: 0.01129819 0.9887018
## class 2: 0.63709223 0.3629078
## class 3: 0.30494914 0.6950509
## class 4: 0.06238342 0.9376166
##
## $Scissors
##           Pr(1)      Pr(2)
## class 1: 0.005780905 0.9942191
## class 2: 0.111216129 0.8887839
## class 3: 0.013239550 0.9867604
## class 4: 0.047631039 0.9523690
##
## $Cactus
##           Pr(1)      Pr(2)
## class 1: 0.02525382 0.9747462
## class 2: 0.45806410 0.5419359
## class 3: 0.06125751 0.9387425
## class 4: 0.29691991 0.7030801
##
## $President
##           Pr(1)      Pr(2)
## class 1: 5.441065e-119 1.0000000
## class 2: 6.064720e-01 0.3935280
## class 3: 3.149983e-02 0.9685002
## class 4: 9.531016e-02 0.9046898
##
## $VP
##           Pr(1)      Pr(2)
## class 1: 0.2405939 7.594061e-01
## class 2: 1.0000000 1.682929e-29
## class 3: 0.5196684 4.803316e-01
## class 4: 0.8702308 1.297692e-01

```

```

lca_ready$.rowid <- seq_len(nrow(lca_ready))
mf <- model.frame(f_ind, data = lca_ready, na.action = na.omit)
idx <- as.integer(row.names(mf))
tab <- table(lca_k4$predclass, lca_ready$wave[idx])
tab

```

```

##
##      W13
##  1 11909
##  2   241
##  3   431
##  4  1119

```

With covariates

```
set.seed(687)
f_cov <- cbind(Month, Day, Year, DOW, Scissors, Cactus, President, VP) ~
  age + factor(gender) + factor(degree) + factor(race) + factor(usborn) + factor(hispanic)

lca_cov <- polLCA(f_cov, data = lca_ready, nclass = 4, maxiter = 5000, nrep = 20, verbose = TRUE)

## Model 1: llik = -25503.41 ... best llik = -25503.41
## Model 2: llik = -28842.3 ... best llik = -25503.41
## Model 3: llik = -25507.15 ... best llik = -25503.41
## Model 4: llik = -28624.86 ... best llik = -25503.41
## Model 5: llik = -26603.06 ... best llik = -25503.41
## Model 6: llik = -26656.55 ... best llik = -25503.41
## Model 7: llik = -26693.14 ... best llik = -25503.41
## Model 8: llik = -26948.94 ... best llik = -25503.41
## Model 9: llik = -26645.44 ... best llik = -25503.41
## Model 10: llik = -26682.8 ... best llik = -25503.41
## Model 11: llik = -26697.04 ... best llik = -25503.41
## Model 12: llik = -26615.2 ... best llik = -25503.41
## Model 13: llik = -26635.48 ... best llik = -25503.41
## Model 14: llik = -26086.59 ... best llik = -25503.41
## Model 15: llik = -26635.93 ... best llik = -25503.41
## Model 16: llik = -29445.39 ... best llik = -25503.41
## Model 17: llik = -25485.47 ... best llik = -25485.47
## Model 18: llik = -26636.17 ... best llik = -25485.47
## Model 19: llik = -27307.02 ... best llik = -25485.47
## Model 20: llik = -27411.22 ... best llik = -25485.47
## Conditional item response (column) probabilities,
## by outcome variable, for each class (row)
##
## $Month
##           Pr(1) Pr(2)
## class 1: 0.0281 0.9719
## class 2: 0.4946 0.5054
## class 3: 0.0146 0.9854
## class 4: 0.0185 0.9815
##
## $Day
##           Pr(1) Pr(2)
## class 1: 0.2002 0.7998
## class 2: 0.8576 0.1424
## class 3: 0.1447 0.8553
## class 4: 0.0485 0.9515
##
## $Year
##           Pr(1) Pr(2)
## class 1: 0.0299 0.9701
## class 2: 0.5845 0.4155
## class 3: 0.0086 0.9914
## class 4: 0.0093 0.9907
##
## $DOW
##           Pr(1) Pr(2)
```

```

## class 1: 0.0415 0.9585
## class 2: 0.4441 0.5559
## class 3: 0.0145 0.9855
## class 4: 0.0120 0.9880
##
## $Scissors
##           Pr(1) Pr(2)
## class 1: 0.0325 0.9675
## class 2: 0.0658 0.9342
## class 3: 0.0046 0.9954
## class 4: 0.0000 1.0000
##
## $Cactus
##           Pr(1) Pr(2)
## class 1: 0.2197 0.7803
## class 2: 0.2890 0.7110
## class 3: 0.0098 0.9902
## class 4: 0.0000 1.0000
##
## $President
##           Pr(1) Pr(2)
## class 1: 0.0377 0.9623
## class 2: 0.3504 0.6496
## class 3: 0.0017 0.9983
## class 4: 0.0000 1.0000
##
## $VP
##           Pr(1) Pr(2)
## class 1: 0.7840 0.2160
## class 2: 0.8484 0.1516
## class 3: 0.1672 0.8328
## class 4: 0.3267 0.6733
##
## Estimated class population shares
## 0.2702 0.0481 0.6609 0.0208
##
## Predicted class memberships (by modal posterior prob.)
## 0.2478 0.0436 0.6887 0.0199
##
## =====
## Fit for 4 latent classes:
## =====
## 2 / 1
##           Coefficient Std. error t value Pr(>|t|)
## (Intercept) -10.64105 0.59720 -17.818 0.000
## age 0.12845 0.00651 19.739 0.000
## factor(gender)2 -0.43957 0.12261 -3.585 0.000
## factor(degree)1 0.07003 0.25805 0.271 0.786
## factor(degree)2 0.15999 0.14789 1.082 0.281
## factor(degree)3 0.88915 0.38743 2.295 0.023
## factor(degree)4 1.40872 0.29762 4.733 0.000
## factor(degree)5 1.84696 0.37422 4.935 0.000
## factor(degree)6 1.39433 0.63119 2.209 0.029
## factor(degree)9 1.04379 0.53219 1.961 0.052

```

```

## factor(race)1      1.70612      0.23926      7.131      0.000
## factor(race)2      1.20649      0.23175      5.206      0.000
## factor(race)7      1.54694      0.22965      6.736      0.000
## factor(usborn)5    -0.95265      0.19192     -4.964      0.000
## factor(usborn)9    -3.73786      0.07157    -52.227      0.000
## factor(hispanic)1  -1.78348      0.69268     -2.575      0.011
## factor(hispanic)2  -2.12237      0.72271     -2.937      0.004
## factor(hispanic)3  -9.50667      0.00332  -2865.913      0.000
## factor(hispanic)5  -1.90136      0.69872     -2.721      0.007
## =====
## 3 / 1
##               Coefficient   Std. error   t value   Pr(>|t|)
## (Intercept)      -6.75778      0.79583    -8.492      0.000
## age               0.01318      0.00386     3.410      0.001
## factor(gender)2   -0.32189      0.08591    -3.747      0.000
## factor(degree)1    1.12917      0.17768     6.355      0.000
## factor(degree)2    1.85935      0.12071    15.404      0.000
## factor(degree)3    2.70581      0.28523     9.486      0.000
## factor(degree)4    3.71276      0.23959    15.496      0.000
## factor(degree)5    4.12755      0.30087    13.719      0.000
## factor(degree)6    3.91239      0.48597     8.051      0.000
## factor(degree)9    2.80263      0.20095    13.947      0.000
## factor(race)1      1.98580      0.93490     2.124      0.035
## factor(race)2      0.55068      0.93331     0.590      0.556
## factor(race)7      0.93100      0.93543     0.995      0.321
## factor(usborn)5    -1.68815      0.13743   -12.284      0.000
## factor(usborn)9    -2.82841      1.94412    -1.455      0.148
## factor(hispanic)1  2.95285      0.29718     9.936      0.000
## factor(hispanic)2  3.04284      0.30741     9.898      0.000
## factor(hispanic)3  4.33322      0.76845     5.639      0.000
## factor(hispanic)5  4.06512      0.27891    14.575      0.000
## =====
## 4 / 1
##               Coefficient   Std. error   t value   Pr(>|t|)
## (Intercept)       3.96828      2.01850     1.966      0.051
## age              -0.18335      0.10004    -1.833      0.069
## factor(gender)2    3.37310      1.92310     1.754      0.081
## factor(degree)1   -0.03916      0.00884    -4.429      0.000
## factor(degree)2   -3.84354      0.00822  -467.791      0.000
## factor(degree)3   12.18695      1.06783    11.413      0.000
## factor(degree)4    2.91335      0.02745   106.145      0.000
## factor(degree)5    3.60086      0.01688   213.374      0.000
## factor(degree)6    5.52622      0.06496    85.066      0.000
## factor(degree)9   -3.00038      0.00003 -92881.507      0.000
## factor(race)1     -3.59063      1.41237     -2.542      0.012
## factor(race)2     -5.65049      1.00307     -5.633      0.000
## factor(race)7    -13.54308      0.03865   -350.412      0.000
## factor(usborn)5   -9.90996      0.04393   -225.605      0.000
## factor(usborn)9    1.73911      0.00000 1906081.188      0.000
## factor(hispanic)1 -8.13449      0.00605  -1345.206      0.000
## factor(hispanic)2  1.72416      2.09138     0.824      0.411
## factor(hispanic)3  4.50547      0.00282   1595.100      0.000
## factor(hispanic)5 -0.67129      1.93214     -0.347      0.729
## =====

```

```
## number of observations: 13700
## number of estimated parameters: 89
## residual degrees of freedom: 166
## maximum log-likelihood: -25485.47
##
## AIC(4): 51148.94
## BIC(4): 51818.68
## X^2(4): 879.4099 (Chi-square goodness of fit)
##
## ALERT: estimation algorithm automatically restarted with new initial values
##
```

```
lca_cov$P
```

```
## [1] 0.27019863 0.04810354 0.66088857 0.02080925
```

```
lca_cov$coeff
```

```
##           [,1]      [,2]      [,3]
## (Intercept) -10.64105405 -6.7577828  3.9682798
## age          0.12845491  0.0131769 -0.1833530
## factor(gender)2 -0.43957099 -0.3218926  3.3731043
## factor(degree)1  0.07002888  1.1291710 -0.0391576
## factor(degree)2  0.15999261  1.8593452 -3.8435371
## factor(degree)3  0.88914609  2.7058090 12.1869481
## factor(degree)4  1.40872467  3.7127626  2.9133532
## factor(degree)5  1.84696223  4.1275546  3.6008615
## factor(degree)6  1.39433303  3.9123924  5.5262198
## factor(degree)9  1.04378720  2.8026322 -3.0003841
## factor(race)1    1.70612359  1.9858009 -3.5906264
## factor(race)2    1.20649457  0.5506784 -5.6504861
## factor(race)7    1.54693799  0.9310027 -13.5430829
## factor(usborn)5 -0.95264862 -1.6881504 -9.9099559
## factor(usborn)9 -3.73785896 -2.8284095  1.7391128
## factor(hispanic)1 -1.78348204  2.9528508 -8.1344926
## factor(hispanic)2 -2.12236755  3.0428434  1.7241623
## factor(hispanic)3 -9.50667162  4.3332164  4.5054650
## factor(hispanic)5 -1.90136125  4.0651242 -0.6712906
```

```
# Without covariates
```

```
saveRDS(lca_k4, "lca_wl3_k4_nocov.rds")
```

```
# With covariates
```

```
saveRDS(lca_cov, "lca_wl3_k4_cov.rds")
```